

PHARMATRACK

Unified Attendance Management & Verification System

University of San Agustin — College of Pharmacy

1. System Description

PharmaTrack is an enterprise-grade attendance verification platform custom-built for the University of San Agustin (USA) College of Pharmacy. The platform streamlines student, facilitator, and administrator workflows during departmental events, classroom lectures, and laboratory sessions.

By replacing traditional paper sheets with high-security static and dynamic QR code verification, PharmaTrack ensures data integrity, eliminates manual reporting delays, and provides stakeholders with instant real-time visualization of attendance analytics.

2. Target User Personas

Pharmacy Students

Students interact with the portal to download their static access pass QR codes, scan facilitator-generated event QR codes, view upcoming calendars, and track their attendance history and standing.

Facilitators (Council & Instructors)

Council officers and instructors configure events, generate classroom QR codes, manage rosters, scan student static QR passes, and export attendance records (PDF/Excel).

System Administrators

Superusers oversee the entire system, manage user registration approvals/rejections, configure global variables (late thresholds, registration mode), and audit operations.

3. Core Features

- **Secure Access Pass:** Each student is assigned a permanent, unique QR Code ID based on their student number, which acts as their digital ID badge.
- **Self-Checkin Scan:** Students can scan dynamic classroom QR codes generated by facilitators to mark themselves present instantly.
- **Full-Screen Presentation Modal:** A dedicated modal on the student dashboard displays their QR pass in high brightness for rapid scanning by council members.
- **Automated Statusing:** System automatically categorizes arrival times into Present, Late, or Absent based on rules configured per event.
- **Self-Profile Repair:** Students with incomplete setup profiles can trigger an inline form to verify and regenerate their QR credentials safely.

4. User Interfaces

Student Dashboard

Contains an overall attendance ring chart (donut), present/late/absent counters, upcoming event highlights, quick action buttons, and full-screen QR pass presenter modal.

Facilitator Panel

Features event scheduler form, real-time camera scanner to record attendance, list of students, and filters to export metrics.

Admin Panel

Contains student & facilitator registration logs with approval/rejection actions, event audits, and configuration settings (thresholds, modes).

5. Attendance Logging Rules

Each scheduled event maintains three timestamp parameters: Check-in Starts, Mark Late At, and Check-in Ends. Attendance status is mapped automatically as follows:

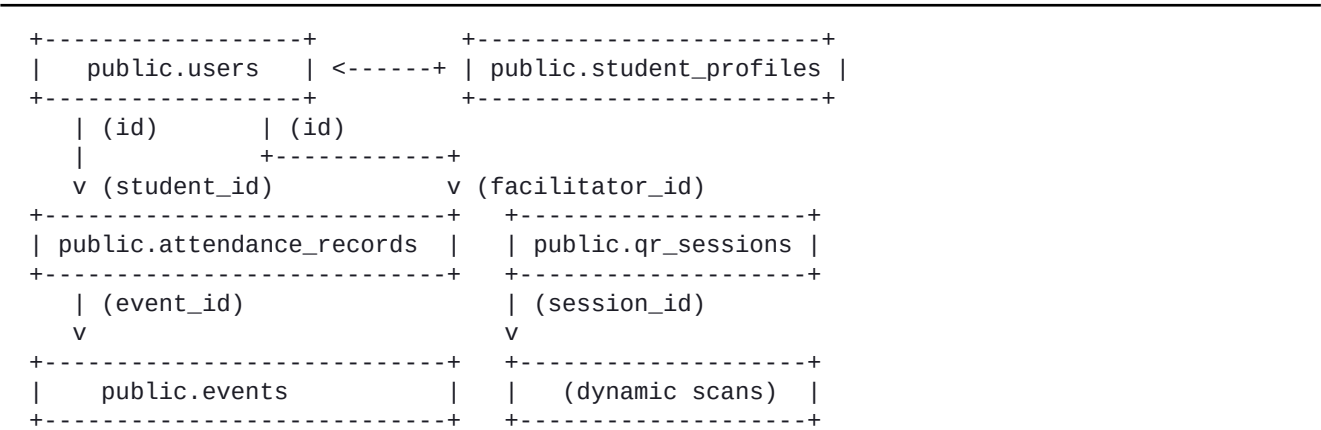
- Check-in Starts $\leq$ Arrival $<$ Mark Late At $\Rightarrow$ **PRESENT**
- Mark Late At $\leq$ Arrival $<$ Check-in Ends $\Rightarrow$ **LATE**
- Arrival $\geq$ Check-in Ends OR No Scan Record $\Rightarrow$ **ABSENT**

6. Data & Schema Storage

PharmaTrack is backed by a Supabase PostgreSQL schema composed of the following tables:

- public.users:** Stores authentication records, account types (student, facilitator, admin), and registration status (pending, approved).
- public.student_profiles:** Stores student metadata (ID number, year, section) and their unique permanent QR identifier.
- public.facilitator_profiles:** Stores faculty/officer profiles (department details).
- public.events:** Configures school activities, start/late/end timelines, locations, and creators.
- public.qr_sessions:** Configures dynamic classroom scan links created by facilitators.
- public.attendance_records:** Logs scanned attendance status, check-in timestamps, scanner identity, and comments.

7. Database Relationships



8. Primary Attendance Flow

Flow A: Facilitator Scans Student Pass (Static)

1. Facilitator logs in -> clicks "Scan Student" -> starts webcam camera.
2. Student logs in -> clicks "Check In" -> opens full-screen high-brightness QR.
3. Facilitator webcam captures student QR ID -> sends payload to Supabase.
4. RLS verifies facilitator is council member -> records present/late logs.

Flow B: Student Scans Session QR (Dynamic)

1. Facilitator starts session -> generates dynamic QR expiring in 10 minutes.
2. Student clicks "Go to Scanner" -> logs in -> opens webcam scan screen.
3. Student scans facilitator session code -> Supabase records class check-in.

